

In this final reflection of the class I would like to look back on my time procuring this system. I would do many things differently. For one, I probably would not have tried to make my system more complex than it had to be. I had implemented the Inter task communication successfully and even had a working demo. But I thought it was immensely to lame to be an effective use of my course knowledge so I expanded, which heavily cost me resources in preparing for other classes.

Content wise, I would reduce the maintenance task time. In my system, the RMS was barely schedulable under the Liu Layland criteria, making the system very harsh to fully note if there were restrictions on implementability especially for larger system architectures with multiple pilots using the same network access to communicate kernel data to each processor. I would revisit making the system more readily understandable with code breakdowns and labels for every function and peripherals such as buttons, LEDs, and even code in the LCD Display but resources are finite at this level of the semester so I'm working with what I got. I would also revisit maybe trying to get the wokwi webpage to work as it seems like a cool feature to make future projects using this kind of thought process and architecture a lot more interesting especially from an educational point of view to get more people to understand the code and the system involvements for scalability.

What was harder than expected was the amount of finagling I had to do to such tiny and minute portions of the code like size the 2048 bit rate to 4096 to prevent stack overflows over the channels. This along with other encountered issues like how to resolve the issue of wanting to create an override button that also still allowed for me to fully demo priority inversion and priority inheritance solutions. If one recalls, this was fully mediated by using the serial reading of the action as the priority blocked task that was being blocked by the maintenance tasking. In this way, the stall emergency can still automatically trigger the runway being clear but may not show on the pilots serial Data monitor. This is irrelevant and is still fully safe as the cockpit indicator would show this transition by allowing the LED to flash 3 times quicker.

Invaluable to me was mostly the struggle. This is where the true learning as an engineer occurs and I'm glad for the hard times. It keeps one thinking of how to overcome these difficult challenges put forth by today's society. The engineers of today will no doubt have the job of reversing many bad decisions and it's only a matter of time before those that didn't struggle will be shown the way of the world when a challenge comes up that costs not a grade, but a human life.